

Python Assignment 1: Data Structures - Strings & Tuples

1. String Concatenation:

```
[13]: string1 = "Hello"
      string2 = input("Enter your name:")
      x = string1 + " " + string2
      print(x)
      string3 = "welcome to python programming"
      print(x + ", " + " " + string3)
```

```
Enter your name: bharath
Hello bharath
Hello bharath, welcome to python programming
```

2. String Slicing and Indexing:

```
[37]: string = "Hello bharath, welcome to python programming"
      print(string[0])
      print(string[-1])
      print(string[0:5])
      print(string[-11:])
      print(string[::-1])
      start = string.find("python")
      print(string[start:start+6])
```

```
H
e
Hello
programming
gnimmargorp nohtyp ot emoclew ,htarahb olleH
python
```

3. String Methods:

```
[47]: strM = "Python beginner tutorial"
      print(strM.upper())
      print(strM.lower())
      print(strM.capitalize())
      print(strM.count('t'))
      print(strM.replace('Python', 'Machine Learning'))
```

```
PYTHON BEGINNER TUTORIAL
python beginner tutorial
Python beginner tutorial
3
Machine Learning beginner tutorial
```

1:

Tuples (Creation, Modification and Access) :

```
[1]: tuple1 = (10, 20, 30)
      tuple2 = (40, 50, 60)
      t_combine = tuple1 + tuple2
      print(t_combine)
      print(t_combine * 3)
      print(t_combine[2])
      print(t_combine[0:3])
      print(t_combine[-3:])

(10, 20, 30, 40, 50, 60)
(10, 20, 30, 40, 50, 60, 10, 20, 30, 40, 50, 60, 10, 20, 30, 40, 50, 60)
30
(10, 20, 30)
(40, 50, 60)
```